Select Kata

The goal of this kata is to practice usage of select.
We will use CTEs to provide the proper data for each case.
That allows creating small cases for quick testing and feedback.

For example

```
with
Animals as (select "Dog" as animal_type
union select "Cat"
union select "Penguin")
Select *
from
Animals
;
```

1.

Complete the following query so you will select only the animal type, despite the existence of two columns.

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cat",  "Mamal"
union select "Penguin", "Bird" )
Select  # Complete here
from
Animals
;
```

2.

Complete the following query so you will select both two columns, **by specifying them**.

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cat",  "Mamal"
```

```
union select "Penguin", "Bird" )
Select  # Complete here
from
Animals
;
```

3.

Complete the following query so you will select the animals **ordered by animal group**.

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cat",  "Mamal"
union select "Penguin", "Bird" )
Select  # Complete here
from
Animals
;
```

4.

Complete the following query so you will select the animals ordered by animal group **and then the animal type**.

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cat",  "Mamal"
union select "Penguin", "Bird" )
Select  # Complete here
from
Animals
;
```

5.

Complete the following query so you will select the animals ordered by animal group and then the animal type **by descending order**.

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cat",  "Mamal"
```

```
union select "Penguin", "Bird" )
Select  # Complete here
from
Animals
;
```

6.

Complete the following query so you will select the **first two animals** ordered by animal group and then the animal type by descending order.

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cat",  "Mamal"
union select "Penguin", "Bird" )
Select  # Complete here
from
Animals
;
```

7.

Complete the following query so you will select the **second animal, when** ordered by animal group and then the animal type by descending order.

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cat",  "Mamal"
union select "Penguin", "Bird" )
Select  # Complete here
from
Animals
;
```

8.

Complete the following query so you will select the **mammals.**

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cat",  "Mamal"
union select "Penguin", "Bird" )
Select  # Complete here
from
Animals
;
```

9.

Complete the following query so you will select the animal **that have 'a'** in their name**. How do you treat case sensitivity?**

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cat",  "Mamal"
union select "Penguin", "Bird"
union select  "Eagle","Bird") # New animal
Select  # Complete here
from
Animals
;
```

10.

Complete the following query so you will select the animal **that have 'a'** in their name or of type "Bird"**.**

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cat",  "Mamal"
union select "Penguin", "Bird"
union select  "Eagle","Bird") # New animal
Select  # Complete here
from
Animals
;
```

11.

Add two new animals. Write a query that selects only one of them.

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cat",  "Mamal"
union select "Penguin", "Bird"
union select  "Eagle","Bird") # New animal
Select  # Complete here
from
Animals
;
```

12.

Add a new column. Write a query that filters the rows using it. **What columns should the query present?**

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cat",  "Mamal"
union select "Penguin", "Bird"
union select  "Eagle","Bird") # New animal
Select  # Complete here
from
Animals
;
```

13.

Select all animals whose name is longer than 3 letters.

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cat",  "Mamal"
union select "Penguin", "Bird"
union select  "Eagle","Bird") # New animal
Select  # Complete here
from
Animals
;
```

14.

Select only cat and eagle, using the “in” operator.

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cat",  "Mamal"
union select "Penguin", "Bird"
union select  "Eagle","Bird") # New animal
Select  # Complete here
from
Animals
;
```

15.

In the following CTE, the name of cat is misspelled.
Who is responsible for the validity?
Who is influenced by the typo?
What are the implications?

Suggest a way to identify typos in animal names.
What are the advantages and disadvantages of your method?

```
with
Animals as (select "Dog" as animal_type, "Mamal" as Animal_Group
union select "Cut",  "Mamal" # Added a typo
union select "Penguin", "Bird"
union select  "Eagle","Bird") # New animal
Select  # Complete here
from
Animals
;
```